

JavaScript Template Literals Assignment

Section 1: Basic Template Literals and Multi-line Strings

Question 1.1: What will be the output of the following code?

javascript

```
let message = `Hello World`;  
console.log(message);
```

Question 1.2: What will be the output?

javascript

```
let poem = `Roses are red  
Violets are blue  
JavaScript is fun  
And so are you`;  
console.log(poem);
```

Question 1.3: What will be the output?

javascript

```
let info = `This is line 1  
This is line 2  
This is line 3`;  
console.log(info);
```

Section 2: Escape Characters vs Template Literals

Question 2.1: What will be the output of these two approaches?

javascript

```
let quote1 = 'He said, "I\'m learning JavaScript";  
let quote2 = "She replied, \\'That\'s great!\'";  
console.log(quote1);  
console.log(quote2);
```

Question 2.2: What will be the output using template literals?

javascript

```
let quote3 = `He said, "I'm learning JavaScript"`;  
let quote4 = `She replied, "That's great!"`;  
console.log(quote3);  
console.log(quote4);
```

Question 2.3: What will be the output?

javascript

```
let path1 = 'C:\\Users\\Documents\\file.txt';  
let path2 = `C:\\Users\\Documents\\file.txt`;  
console.log(path1);  
console.log(path2);
```

Section 3: Template Literals with Placeholders

Question 3.1: What will be the output?

javascript

```
let username = 'Alice';  
let score = 95;  
console.log('User: ' + username + ', Score: ' + score);  
console.log(`User: ${username}, Score: ${score}`);
```

Question 3.2: What will be the output?

javascript

```
let product = 'Laptop';  
let price = 1200;  
console.log(`The ${product} costs ${price}`);
```

Question 3.3: What will be the output?

javascript

```
let firstName = 'John';  
let lastName = 'Doe';  
console.log(`Full Name: ${firstName} ${lastName}`);
```

Section 4: Expressions in Template Literals

Question 4.1: What will be the output?

javascript

```
let x = 15;  
let y = 25;  
console.log(`The sum of ${x} and ${y} is ${x + y}`);
```

Question 4.2: What will be the output?

javascript

```
let length = 10;  
let width = 5;  
console.log(`Area: ${length * width}`);  
console.log(`Perimeter: ${2 * (length + width)}`);
```

Question 4.3: What will be the output?

javascript

```
let num1 = 50;  
let num2 = 30;  
console.log(`${num1} + ${num2} = ${num1 + num2}`);  
console.log(`${num1} - ${num2} = ${num1 - num2}`);
```

Section 5: Complex Template Literals

Question 5.1: What will be the output?

javascript

```
let item = 'Book';  
let quantity = 3;  
let unitPrice = 25;  
console.log(`Item: ${item}  
Quantity: ${quantity}  
Unit Price: ${unitPrice}  
Total: ${quantity * unitPrice}`);
```

Question 5.2: What will be the output?

javascript

```
let name = 'Sarah';
let age = 28;
let city = 'New York';
console.log(`Hello, my name is ${name}.
I am ${age} years old.
I live in ${city}.`);
```

Question 5.3: What will be the output?

javascript

```
let a = 8;
let b = 12;
let c = 20;
console.log(`Numbers: ${a}, ${b}, ${c}`);
console.log(`Average: ${ (a + b + c) / 3 }`);
```

Section 6: Comparison with String Concatenation

Question 6.1: What will be the output of both approaches?

javascript

```
let courseName = 'JavaScript';
let duration = 40;
let instructor = 'Naveen';
```

// Traditional concatenation

```
console.log('Course: ' + courseName + ', Duration: ' + duration + ' hours, Instructor: ' + instructor);
```

// Template literals

```
console.log(`Course: ${courseName}, Duration: ${duration} hours, Instructor: ${instructor}`);
```

Question 6.2: What will be the output?

javascript

```
let browserName = 'Chrome';
let version = 91;
console.log(`Browser: ${browserName} ${version}`);
console.log('Browser: ' + browserName + ' ' + version);
```

Section 7: Nested Template Literals

Question 7.1: What will be the output?

javascript

```
let isLoggedIn = true;
let username = 'testuser';
console.log(`Status: ${isLoggedIn ? `Welcome ${username}` : 'Please login'}`);
```

Question 7.2: What will be the output?

javascript

```
let temperature = 25;
console.log(`Weather: ${temperature > 30 ? 'Hot' : temperature > 20 ? 'Warm' : 'Cold'}`);
console.log(`Temperature: ${temperature}°C`);
```

Section 8: Mixed Operations

Question 8.1: What will be the output?

javascript

```
let testName = 'Login Test';
let status = 'Pass';
let executionTime = 2.5;
console.log(`Test: ${testName}
Status: ${status}
Execution Time: ${executionTime} seconds`);
```

Question 8.2: What will be the output?

javascript

```
let baseUrl = 'https://example.com';
let endpoint = '/api/users';
let userId = 123;
console.log(`Full URL: ${baseUrl}${endpoint}/${userId}`);
```